

Predicting Customer Churn in a Digital Financial Services Platform: A Machine Learning Classification Approach

Data Science Analytics Assignment — Samba Balde 3520872

Tools: Microsoft Excel | Orange Data Mining 3.x | Dataset: 5,009 customers across France, Germany & Spain | Target: Left_Service (Binary Classification)

20.6%

CHURN RATE

5,009

CUSTOMERS

AUC 0.846

BEST MODEL

INTRODUCTION & BACKGROUND

Customer churn — the loss of clients to competitors or disengagement — represents a critical risk for digital financial services platforms. Acquiring new customers costs five to seven times more than retaining existing ones, making early churn prediction a commercially significant analytical challenge.

This study applies machine learning classification to a dataset of 5,009 anonymised customers to identify which variables most strongly predict churn, and to evaluate the comparative performance of three classification algorithms: Random Forest, Logistic Regression, and Naïve Bayes.

RESEARCH QUESTIONS

RQ1: Which customer attributes are the strongest predictors of churn in a digital financial services context?

RQ2: Which classification algorithm achieves the highest discriminatory performance on this dataset?

RQ3: Are there statistically significant differences in demographic or financial profile between churned and retained customers?

DATA & METHODOLOGY

Three CSV datasets were merged in Microsoft Excel using **Client_ID** as the join key via VLOOKUP, producing a master dataset of 14 variables.

- Import & Merge:** Three CSVs joined on Client_ID via VLOOKUP in Excel
- Deduplication:** 3 duplicate Client_IDs removed; 5,009 clean records retained
- Missing Value Imputation:** Mode used for binary variables; median for Annual_Earnings (continuous)
- Zero-Variance Exclusion:** Premium_Tier excluded (all values = 0; no predictive contribution)
- Train/Test Split:** 80/20 stratified split (4,008 training / 1,001 test)
- Classification:** Random Forest, Logistic Regression, Naïve Bayes evaluated in Orange

DATASET OVERVIEW

Total customers	5,009
Churned (Left_Service = 1)	1,034 (20.6%)
Retained (Left_Service = 0)	3,975 (79.4%)
Features used in modelling	10
Features excluded	3 (ID, Name, Zero-var)
Missing values (post-imputation)	0
Regions	France, Germany, Spain
Train / Test split	4,008 / 1,001
Stratified sampling	Yes

VARIABLE TYPES

Numeric Age, Account_Value, Membership_Length, Risk_Score, Annual_Earnings, Product_Count

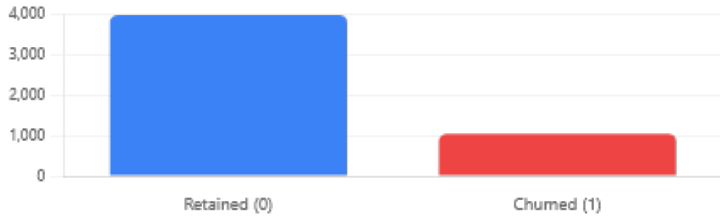
Categorical Gender, Region, CreditCard_Flag, Activity_Status

Target Left_Service (0 = retained, 1 = churned)

DESCRIPTIVE ANALYSIS

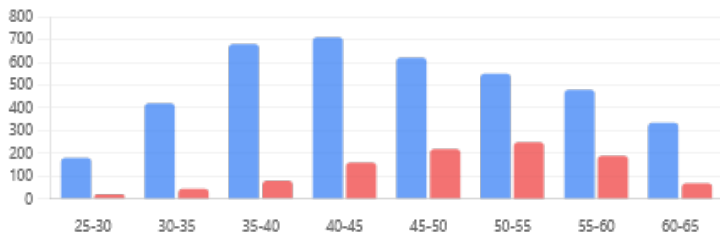
Churn Distribution (Left_Service)

Retained (0) — 79.4% | Churned (1) — 20.6%



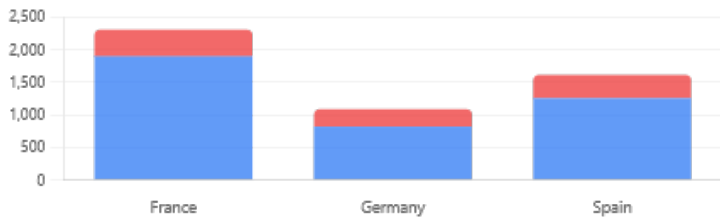
Age Distribution by Churn Status

Retained: mean 37.3yrs | Churned: mean 44.8yrs



Student's t = 21.84 (p < 0.001): Age difference is **highly statistically significant**. Churners are on average **7.5 years older** than retained customers.

Churn Rate by Region



Germany exhibits the highest churn proportion relative to its customer base, visible in the Mosaic Display analysis.

Account Value by Churn Status (mean ± SD)



Student's t = 1.689 (p = 0.091): Account value difference is **not statistically significant** (p > 0.05). Account value alone does not predict churn.

FEATURE IMPORTANCE (INFORMATION GAIN RATIO & GINI)

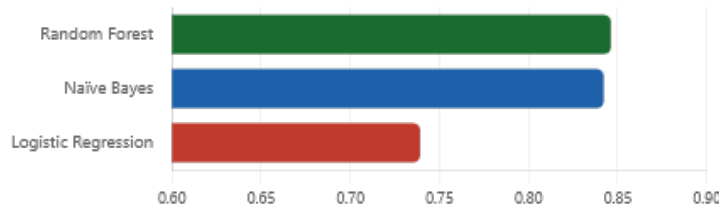
Ranked using the Orange Rank widget with Information Gain Ratio and Gini Decrease scoring methods.

Variable	Gain Ratio (scaled)	Score
Product_Count	#1	0.085
Age	#2	0.043
Activity_Status		0.014
Region		0.013
Gender		0.009
Account_Value		0.007
Membership_Length		0.000
Risk_Score		0.000
Annual_Earnings		0.000

Key finding: Product_Count (Gain Ratio = 0.085) and Age (0.043) are the dominant predictors of churn — far outweighing financial variables. Risk_Score, Annual_Earnings, and Membership_Length carry zero discriminatory value in this dataset.

MODEL EVALUATION — TEST AND SCORE (N=1,001)

AUC Comparison — All Three Models



Full Performance Metrics

Model	AUC	CA	F1	Prec.	Recall	MCC
Random Forest ★	0.846	0.860	0.844	0.853	0.860	0.521
Naïve Bayes	0.842	0.836	0.821	0.821	0.836	0.437
Logistic Regression	0.739	0.802	0.761	0.767	0.802	0.241

★ Best performing model across all metrics. Evaluation method: test on held-out data (n=1,001, stratified 80/20 split).

CONFUSION MATRIX — RANDOM FOREST (TEST SET, N=1,001)

Actual Class ↑	Predicted Class →		
0	0	769	True Negative Correctly identified as retained
0	1	25	False Positive Predicted churn; actually retained
1	0	115	False Negative Δ Missed churners (most costly error)
1	1	92	True Positive Correctly identified churners
		78.6%	CHURN PRECISION
		44.4%	CHURN RECALL

The model correctly identifies only 44.4% of actual churners (92/207), suggesting class imbalance (20.6% churn vs. 79.4% retained) limits minority-class detection.

KEY FINDINGS & DISCUSSION

F1. Product_Count is the single strongest churn predictor (Gain Ratio = 0.085), suggesting customers with fewer products have less platform integration and are more likely to disengage.

F2. Age is a highly significant discriminator (t = 21.84, p < 0.001). Churned customers are on average 7.5 years older (mean 44.8 vs. 37.3), pointing to potential engagement or UX gaps for older demographics.

F3. Account value does NOT significantly differ between groups (p = 0.091), challenging the assumption that high-value customers are more loyal.

F4. Random Forest achieves the best overall performance (AUC = 0.846, MCC = 0.521), though Naïve Bayes performs comparably (AUC = 0.842) at far lower computational cost — a notable finding for resource-constrained deployments.

CONCLUSIONS & LIMITATIONS

This analysis demonstrates that customer churn in digital financial services is predictable with reasonable accuracy using a Random Forest classifier. The most practically actionable findings are that **product cross-selling** and **age-targeted engagement strategies** are the highest-priority interventions for retention teams.

Limitations: Class imbalance (79.4% vs. 20.6%) limits churn-class recall to 44.4%, meaning over half of real churners are missed. Future work should apply SMOTE oversampling or cost-sensitive learning. Premium_Tier's zero variance across all records may indicate a data collection issue worth investigating. The dataset covers three European markets only, limiting generalisability.

TOOLS & REPRODUCIBILITY

Data cleaning & merge	Microsoft Excel (Power Query, VLOOKUP)
ML pipeline & visualisation	Orange Data Mining 3.x
Train/test split	80/20 stratified, fixed random seed
RF configuration	100 trees, replicable training enabled
Feature scoring	Information Gain Ratio + Gini Decrease

